LBNL IC GROUP DESIGN NOTE

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LArPix-v3 Datasheet

The Liquid Argon Pixel readout ASIC, Version 3 (LArPix-v3) is a data acquisition ASIC intended for use with a 2D liquid argon time-project chamber (LAr-TPC). 2D readout is a proposed technology for the DUNE Near Detectors. The datasheet is intended to communicate the functionality and use of LArPix-v3. LArPix-v3 is implemented in 130 nm CMOS technology.

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REVISION NUMBER	KEY CHANGES	DATE	ENGINEER
1.0	Initial draft.	1/23/2023	CRG
2.0	Major update to include details on changes, quality improvements, and new features in LArPix-v3.	4/15/2023	CRG
2.1	Changes to incorporate improved documentation and additional functionality added after design review.	4/27/2023	CRG
2.2	Documented analog changes, padring, and configuration bits and defaults.	1/16/2024	PZ
2.3	Fixed a few errors in documentation.	6/27/2024	CRG
2.4	Updated **adc_ibias_delay** description	12/13/2024	CRG

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1 Document Description

The Datasheet for Version 3 of the Liquid Argon Pixel (LArPix-v3) readout ASIC is intended to communicate the functionality and use of LArPix-v3. Details relevant for applying LArPix-v3 in a system are highlighted.

1 Differences between LArPix-v2c and LArPix-v3

For readers already familiar with LArPix ASICs, the key differences between LArPix-v3 and LArPix-v2c are summarized here.

1.1 Digital Changes

There are significant feature and architectural differences between LArPix-v2c and LArPix-v3.

1.1.1 Digital Correlated Double-Sampling

LArPix-v3 includes an optional correlated double-sampling (CDS) capability. When CDS is enabled, the LArPix-v3 channels will capture samples of the CSA pedestal and include them as independent output packets (the subtraction of pedestal and data must be done off-chip). Using CDS largely eliminates reset and 1/f noise because the low-frequency components of the data and pedestal samples are correlated. However, CDS increases thermal noise (by a factor of $\sqrt{2}$) because the thermal noise of the data and pedestal samples are uncorrelated. Therefore, the choice of whether it makes sense to enable CDS depends on what is limiting the channel noise performance and requirements on data volume. One consequence of adding CDS is that the timestamp is slightly shorter (reduced from 32 bits to 28 bits). The timestamp was shortened because the CDS status needs to be included in the output packet and there was a strong desire to keep the packet size to be 64 bits.

1.1.2 Asynchronous ADC Controller

Previous versions of LArPix included an 8-bit, Successive Approximation (SAR) ADC using a synchronous controller implemented in RTL. The ADC in LArPix-v3 has been significantly upgraded. The new 10-bit ADC includes a new full-custom asynchronous digital controller. Moving the controller out of RTL also allowed simplification of the channel controller as the controller no longer needs to execute the SAR algorithm.

1.1.3 Independent Configuration FIFO

In earlier versions of LArPix, the configuration and data paths shared a common FIFO. While this simplified data handling internal to LArPix, it made it challenging to recover a Hydra when it is experiencing a packet storm. By separating the Configuration and Data FIFOs, it is much easier to disable hot channels or get control of a LArPix-v3 ASIC in case it is generating enough erroneous data to fill its FIFO.

1.1.4 FIFO Usage Statistics Available through Configuration Interface

In previous versions of LArPix, the configuration registers' write path was only exposed to the UART, so it was not possible for LArPix to update the values stored in the configuration on its own. Separating the configuration FIFO from the data FIFO made it possible for the LArPix-v3 core to update configuration registers in a straightforward way. Several diagnostics have been added to LArpix-v3, including total packets written into the FIFO since the last reset, the peak FIFO occupancy since the last reset, and how many packets have been dropped since the last reset. In addition, it is now possible to put the last four bits of the running packet tally in each packet. This may make it easier to detect missed packets.

1.1.5 Magic Number for Configuration Packets

To protect against malformed packets reconfiguring different LArPix ASICs in a Hydra network, configuration packets are now required to include a specific, hardwired 32-bit "magic number" before a configuration command is executed. The new magic number makes it extremely unlikely that random or malformed packets will result in an unintended reconfiguration of a downstream or upstream ASIC.

1.1.6 Changed Data Packet Identifier

In all versions of LArPix, the first two bits of a packet identify the packet type (i.e. data, config read or write, or – in earlier versions - test). In LArPix-v3, the packet identifier for data was changed from "00" to "01". This has the beneficial effect that an all-zero packet is now an invalid packet. This change is an

additional line of defense in case an RX receiver is stuck at zero. Now the packets it generates will not contribute to a “packet storm” that could make a network of LArPix-v3 ASICs unstable.

1.1.7 Capability to Drop Malformed Packets

Earlier versions of LArPix were completely agnostic regarding the contents of packets that weren’t intended for them, or for data packets. In LArPix-v3, the packets are parsed for a valid packet type and a valid magic number (in case of configuration packets) and are dropped if these values are not correct. LArPix-v3 maintains a running total of the number of packets it has dropped since the last time it has been reset. This total dropped packets value is accessible through the configuration interface.

1.1.8 Parity Generated in event_builder

In earlier versions of LArPix, parity (a measure of how many bits are set in a word) was used primarily to protect and study the channel between the leaf LArPix and PACMAN. As is traditional in UART-based communication systems, the parity was generated in the transmit UART and then stripped by the receiver. In other words, it was calculated in the UART in each LArPix chip and no parity information was stored in the FIFO. LArPix users have indicated a desire to use parity to track errors across tiles. To do this, parity is now calculated once, when the packet is created, and the parity must be preserved in the FIFO as the packet is set from chip to chip before it reaches the PACMAN.

1.1.9 Configuration Writes Gated by Parity

In earlier versions of LArPix, parity (a measure of how many bits are set in a word) was used primarily to protect and study the channel between the leaf LArPix and PACMAN. It wasn’t used internally for any purpose. In LArPix-v3, we will only execute a configuration write command if the parity is correct. This protects against possible errors where a random bit switches in the configuration word, which could be only a payload or address. In this case, the magic number would be intact but either data would be written to the wrong address or the wrong data would have been written to the wrong address. The parity bit protects against a single bit flip (an odd number of bit flips, to be precise).

1.1.10 Latch-Based FIFO Memory

The RAM array that implements the FIFO has been converted to a latch-based design. This allows the FIFO size to be increased back to 2048 address locations. LArPix-v1 used a Flip-Flop-based design, LArPix-v2a and LArPix-v2b used SRAM-based designs, and LArPix-v2c also used a Flip-Flop-based design (with a half-sized FIFO compared to other ASICs in the LArPix family).

1.1.11 Enable Bits for External Trigger and External Sync

To isolate the LArPix-v3 digital core from switching in response to external trigger and sync signals, dedicated enable bits have been added to the configuration registers.

1.1.12 Removed Test Modes from UART

In previous versions of LArPix, the UART contained test features for characterizing the quality of a Hydra link. For instance, the UART could send several predetermined patterns independent of its input (PRSB-5, 01, etc.). This functionality was initially included to enable debugging of the communication interface, but this has not been needed, nor has it been used in practice. Therefore, the UART test modes have been removed from LArPix-v3 to simplify the UART.

Because of this change, test packets are no longer defined.

1.2 Analog Changes

1.2.1 New 10-bit Asynchronous ADC

The ADC in LArPix-v3 has been significantly upgraded compared to the ADC included in earlier versions of LArPix. The ADC is now fully differential, uses a more sophisticated switching scheme, and is asynchronous. Taken together, this means the new ADC is faster, lower noise, higher resolution, lower power, and more robust compared to the older ADC.

1.2.2 New Channel Buffer

A new channel buffer is included in LArPix-v3 to drive the ADC input capacitors. Based on a super source-follower architecture, it can achieve fast settling with high current efficiency. Combined with the reduced capacitance of the ADC sampling array the new buffer uses significantly less power than the channel buffer included in earlier versions of LArPix. Differential drive is not yet implemented but is planned for future LArPix versions.

1.2.3 Increased Hydra I/O Maximum Loop Current

In earlier versions of LArPix that included Hydra I/O, the nominal loop current was limited to approximately 225 µA. The maximum single-ended loop current has been increased to approximately 750 µA to allow for more robust interfacing with the PACMAN controller.

1.2.4 Fully Differential Analog Monitor with Line Driving Capability

The Analog Monitor has been significantly upgraded in terms of functionality. It is now fully differential (to reduce noise and interference coupling on the tile) and it now has the ability to drive the tile capacitance. The new Analog Monitor also provides sufficiently good noise and bandwidth performance that it will be possible to use it for in-situ evaluation of the raw CSA outputs prior to digitization. The Analog Monitor and ADC Test pads are now shared.

1.2.5 Differential ADC Test Port

The pads that connect to the ADC test port are fully differential in LArPix-v3. The Analog Monitor and ADC Test pads are now shared. Which path is active (ADC test or Analog Monitor) is determined by a configuration bit (**en_analog_monitor**).

1.2.6 Removed Adjustability of CSA Gain

In previous versions of LArPIx, the gain of the CSA was configurable to different values. This capability has been removed for LArPix-v3.

1.2.7 New Reference Generator for ADC Asynchronous Logic

The new ADC asynchronous logic requires a built-in delay cell to ensure that the capacitor DAC has fully settled before the next bit decision is made by the main ADC comparator. To this end, a voltage ramp is compared to a voltage reference using a discriminator block similar to the one in the channel's AFE. This reference is generated in the main biasing cell (AnalogDACs) using a 2-bit resistive divider. That resistive divider is controlled by two configuration bits (**adc_comp_trim[1:0]**).

1.2.8 New I_{bias} Cell for ADC Asynchronous Logic

The voltage ramp mentioned in Section 2.2.7 above is created by integrating a current on a capacitor. This current is generated through a current bias cell similar to the ones used for other AFE blocks (such as the CSA or Buffer), and placed in all four corners of the chip. This current value, and consequently the delay it generates, is programmable through 4 configuration bits, **adc_ibias_delay[3:0]**. To offer monitor capability of these bias currents through the analog monitor, 4 one-hot configuration bits named **adc_ibias_delay_monitor[3:0]** have been introduced. Lastly, it should be noted that an additional output has been included in the reference current generation block to serve as input to this ADC delay current bias.

2 LArPix-v3 Concept

The LAr Pixel ASIC, Version 3 (LArPix-v3) is a data acquisition ASIC intended for use with pixelated 2-D liquid argon time-project chambers (LAr-TPCs). Two-dimensional, pixelated readout is a proposed technology for the DUNE Near Detectors. The basic idea is to implement a 2D area detector using a PCB with vias as the pixel sensors. Biased strips on the PCB focus electrons onto the vias and the collected charge is read out using LArPix-v3 ASICs.

Each LArPix-v3 ASIC contains 64 analog channels. LArPix-v3 is organized conceptually in quadrants, and the individual channels are distributed around the chip as shown in Figure 1. There are 16 channels on each side as well as a digital I/O block (labeled POSI/PISO) in each corner of the chip. Each channel comprises a charge-sensitive amplifier (CSA), a discriminator with a programmable threshold, and a

10-bit analog-to-digital converter (ADC). Charge incident on the input of each channel is converted by the CSA to a voltage that is proportional to the charge received since the last reset pulse. If this voltage is larger than the threshold, the discriminator fires and an ADC conversion is initiated. The ADC output is then assembled into an event by the digital core. In addition to the ADC value, each event includes the channel and chip IDs as well as a time stamp.

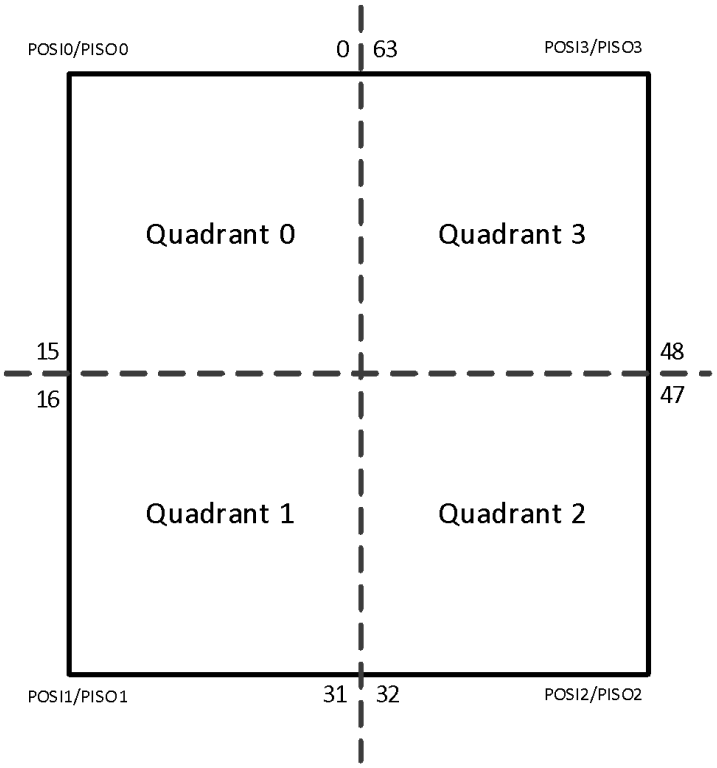


Figure 1. Quadrant Definitions for LArPix-v3.

A block diagram of LArPix-v3 channel allocation is shown in Figure 1. The 64 input channels are organized in four independent banks of 16 channels each. The POSI/PISO connections can be defined as desired by the system designer (as long as the designer is consistent). As stated above, please note that the POSI/PISO quadrants have independent power connections and must all be connected externally.

To reduce the number of connections external to the cryostat, a collection of LArPix-v3 chips are organized as a network with a single LArPix-v3 chip functioning as the interface between the network and the control electronics. This functionality is called “Hydra-I/O” and is described in more detail in its own section.

In addition to data-driven operation, LArPix-v3 also includes programmable external trigger, reset, and cross-trigger functions. In the cross-trigger function, all appropriately configured channels will generate an event if a hit occurs in any channel.

3 LArPix-v3 Pads (Bare Die)

The LArPix-v3 padring is shown in Figure 2. LArPix-v3 has 205 pads arrayed on a 92 µm pitch. Each pad is 96 µm long and 53 µm wide. The pads for the bare die are numbered counter-clockwise with pad #1 the top pad on the left side of the chip. Pad 205 is a dummy pad connected to VSSD. The pads are listed in Table 3.

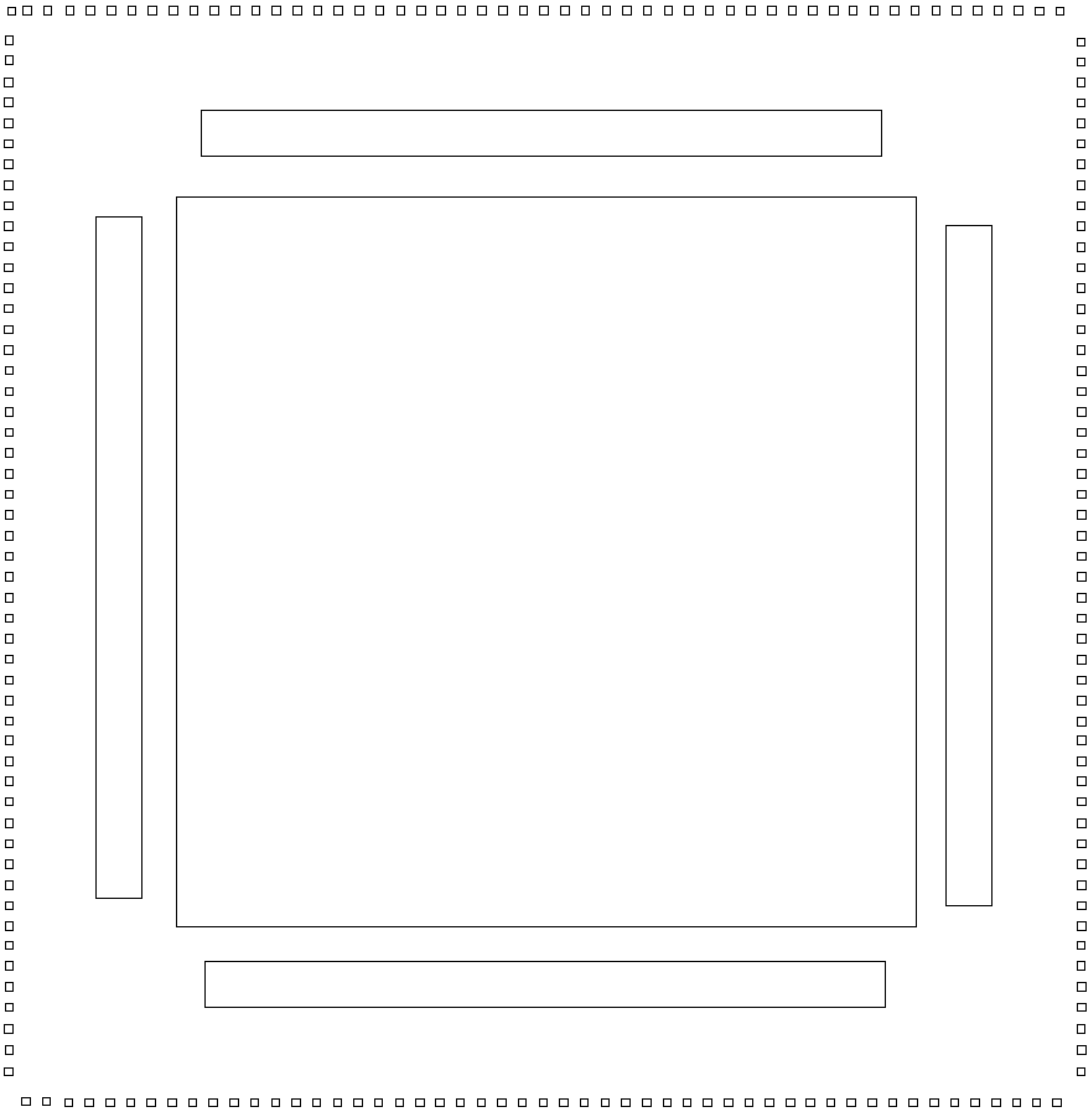


Figure 2. LArPix-v3 padring.

PAD	PAD NAME	PAD	PAD NAME	PAD	PAD NAME	PAD	PAD NAME
1	PISO0_P	52	PISO1_P	103	PISO2_N	154	PISO3_P
2	PISO0_N	53	PISO1_N	104	PISO2_P	155	PISO3_N

3	CLK_P	54	VDDD	105	EXT_TRIG_SYNC _N	156	VSSD
4	CLK_N	55	VSSD	106	EXT_TRIG_SYNC _P	157	VDDD
5	VSSD	56	VSSD	107	VDD_IO_2	158	VSSD
6	VDDD	57	VDDA	108	VSS_IO_2	159	VDDA
7	VDDD	58	VSSA	109	VDDA	160	VSSA
8	VSSD	59	I_MONITOR	110	VBP	161	ADC_MON_N
9	VDDA	60	VSSA	111	VSSA	162	ADC_MON_P
10	VSSA	61	VSSA	112	VSSA	163	VSSA
11	QIN<8>	62	QIN<24>	113	QIN<40>	164	QIN<56>
12	VSSA	63	VSSA	114	VSSA	165	VSSA
13	QIN<9>	64	QIN<25>	115	QIN<41>	166	QIN<57>
14	VSSA	65	VSSA	116	VSSA	167	VSSA
15	QIN<10>	66	QIN<26>	117	QIN<42>	168	QIN<58>
16	VSSA	67	VSSA	118	VSSA	169	VSSA
17	QIN<11>	68	QIN<27>	119	QIN<43>	170	QIN<59>
18	VSSA	69	VSSA	120	VSSA	171	VSSA
19	QIN<12>	70	QIN<28>	121	QIN<44>	172	QIN<60>
20	VSSA	71	VSSA	122	VSSA	173	VSSA
21	QIN<13>	72	QIN<29>	123	QIN<45>	174	QIN<61>
22	VSSA	73	VSSA	124	VSSA	175	VSSA
23	QIN<14>	74	QIN<30>	125	QIN<46>	176	QIN<62>
24	VSSA	75	VSSA	126	VSSA	177	VSSA
25	QIN<15>	76	QIN<31>	127	QIN<47>	178	QIN<63>

26	VSSA	77	VSSA	128	VSSA	179	VSSA
27	QIN<16>	78	QIN<32>	129	QIN<48>	180	QIN<0>
28	VSSA	79	VSSA	130	VSSA	181	VSSA
29	QIN<17>	80	QIN<33>	131	QIN<49>	182	QIN<1>
30	VSSA	81	VSSA	132	VSSA	183	VSSA
31	QIN<18>	82	QIN<34>	133	QIN<50>	184	QIN<2>
32	VSSA	83	VSSA	134	VSSA	185	VSSA
33	QIN<19>	84	QIN<35>	135	QIN<51>	186	QIN<3>
34	VSSA	85	VSSA	136	VSSA	187	VSSA
35	QIN<20>	86	QIN<36>	137	QIN<52>	188	QIN<4>
36	VSSA	87	VSSA	138	VSSA	189	VSSA
37	QIN<21>	88	QIN<37>	139	QIN<53>	190	QIN<5>
38	VSSA	89	VSSA	140	VSSA	191	VSSA
39	QIN<22>	90	QIN<38>	141	QIN<54>	192	QIN<6>
40	VSSA	91	VSSA	142	VSSA	193	VSSA
41	QIN<23>	92	QIN<39>	143	QIN<55>	194	QIN<7>
42	VSSA	93	VCAS	144	VSSA	195	VSSA
43	VDDA	94	VDDA	145	VDDA	196	VREF
44	VCM	95	VSSD	146	VSSD	197	VDDA
45	VSSA	96	VDDD	147	DIG_MONITOR	198	VSSD
46	VDDD	97	VDDD	148	VDDD	199	VSSD
47	VSSD	98	VSSD	149	VDDD	200	VDDD
48	VDD_IO_1	99	RST _N	150	VDD_IO_3	201	VDD_IO_0
49	VSS_IO_1	100	RST _P	151	VSS_IO_3	202	VSS_IO_0

50	POSI1_P	101	POSI2_N	152	POSI3_N	203	POSI0_N
51	POSI1_N	102	POSI2_P	153	POSI3_P	204	POSI0_P
						205	VSSD (NoConn)

Table 2. LArPix-v3 padring.

The function of each pad is described in Table 3, including information on required pad connections. The digital I/O circuits used the VDD_IO_X and VSS_IO_X connections. Please note the difference VDD_IO_X domains (e.g. VDD_IO_0, VDD_IO_1, etc) are not shorted together internally. Therefore, each pad connection must be made for LArPix-v3 to function correctly. The internal digital circuits use the nominally 1.2 V VDDD supply which can be lowered to reduce power dissipation. VDDA can be also be lowered from its nominal level of 2.5 V to reduce power dissipation at the expense of dynamic range.

Pad Name	**Direction**	**Type**	**Description**	**Number of Pads**	**Bypass**
QIN[63:0]	IN	Analog	Charge inputs from sensor	64	
VDDA	INOUT	Power	2.5 V (nominal) Analog Power for Channel	9	0.1 µF in parallel with 1 µF to VSS
VDDD	INOUT	Power	1.2 V (nominal) Digital Power	8	0.1 µF in parallel with 1 µF to VSS
VSSA	INOUT	Power	Analog Ground	70	
VSSD	INOUT	Power	Digital Ground	8	
PISO0[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 0. Differential current-mode digital data output from current chip to other chips or FPGA in Hydra network	2	
PISO1[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 1. Differential current-mode digital data output from current	2	

			chip to other chips or FPGA in Hydra network		
PISO2[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 2. Differential current-mode digital data output from current chip to other chips or FPGA in Hydra network	2	
PISO3[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 3. Differential current-mode digital data output from current chip to other chips or FPGA in Hydra network	2	
POSI0[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 0. Differential, current-mode digital input from other chips or FPGA in Hydra network to current chip. Internally terminated.	2	
POSI1[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 1. Differential, current-mode digital input from other chips or FPGA in Hydra network to current chip. Internally terminated.	2	
POSI2[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 2. Differential, current-mode digital input from other chips or FPGA in Hydra network to current chip. Internally terminated.	2	
POSI3[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 3. Differential, current-mode digital input from other chips or FPGA in Hydra network to	2	

			current chip. Internally terminated.		
CLK_{P/N}	IN	Digital	Differential, current mode digital clock input. Can be internally or externally terminated.	2	
EXTERN_TRIG_ SYNC{P/N}	IN	Digital	Differential, current-mode external trigger to initiate conversion (active high). Also provides timestamp synchronization. Can be internally or externally terminated.	2	
RST_{P/N}	IN	Digital	Differential, current-mode synchronous global reset (active low).	1	
ANA_MONITOR	OUT	Analog	Monitor port for various internal analog signals.	1	
DIG_MONITOR	OUT	Digital	Monitor port for various internal digital signals.	1	
I_MONITOR	OUT	Analog	Monitor port for various internal bias currents. Must be terminated with appropriate resistor to ground (monitor bus is a current source).	1	
VREF	INOUT	Analog	External bypass connection pin for VREF.	2	0.1 µF in parallel with 10 µF to VSS
VCM	INOUT	Analog	External bypass connection pin for VCM.	3	0.1 µF in parallel with 10 µF to VSS
RBIAS	INOUT	Analog	Not used in LArPix-v3	1	

VCAS	INOUT	Analog	Internal node brought out for optional filtering. Nominally a no-connect.	1	
VBP	INOUT	Analog	Internal node brought out for optional filtering. Nominally a no-connect.	1	
ADC_MON_{P/N}	INOUT	Analog	Differential output for analog monitor as well as differential test input for ADC. Configurable using control bits.	2	

Table 3. List of LArPix-v3 pads.

4 LArPix-v3 Package

TODO

5 Digital Electrical Interface

5.1 Clock Definition

LArPix-v3 receives a differential LVDS CMOS system clock called CLK. The clock rate can be adjusted to balance event rate and power dissipation. The clock rate also defines the resolution of the internal timestamp (the timestamp resolution is the reciprocal of the clock rate). The nominal clock speed is 5 MHz.

5.2 UART

LArPix-v3 uses a simple, industry-standard serial protocol to transfer data on- and off-chip. To implement this protocol at low power, LArPix-v3 contains a Universal Asynchronous Receiver/Transmitter (UART). LArPix-v3 sends and receives data in a bit-serial fashion. The input port of the UART is called the

Primary-Out-Secondary-In (POSI), and the transmit port is called the Primary-In-Secondary-Out (PISO). In order to coordinate the data transmission, the UART uses a self-synchronizing start- and stop-bit data format as shown in Figure 3.



Figure 3. UART data format.

Each UART word comprises one leading low start bit, one trailing high stop bit, and 64 data bits. The UART does not implement a parity bit directly but parity is implemented in the event building process. The UART operates as follows. When no data transmission is taking place, the output line of the UART is held high. The start bit of a data transmission sequence is indicated by the output falling low. Then the data is sent out one bit per baud[1]. The UART sends bit [0] of the data word first and bit [53] is sent last. After all 64 data bits are sent, the stop bit is indicated by the output going high. The output is then kept high until the next start bit indicates transmission of a new data word. The contents of the UART word are described in the section on the LArPix-v3 configuration interface.

The transmit clock runs at the same frequency as the system clock. This obviates the need for multiple clock domains in the ASIC (which complicated the implementation of earlier versions of LArPix). The key tradeoff here is that the data is not oversampled, but instead the data is launched and captured by the PISO and POSI ports, respectively, on opposite clock edges, as illustrated in Figure 4.

[1] The baud rate in LArPix-v3 is equal to the system clock rate (in contrast to earlier version of LArPix, when the baud rate is nominally the system clock divided by two).

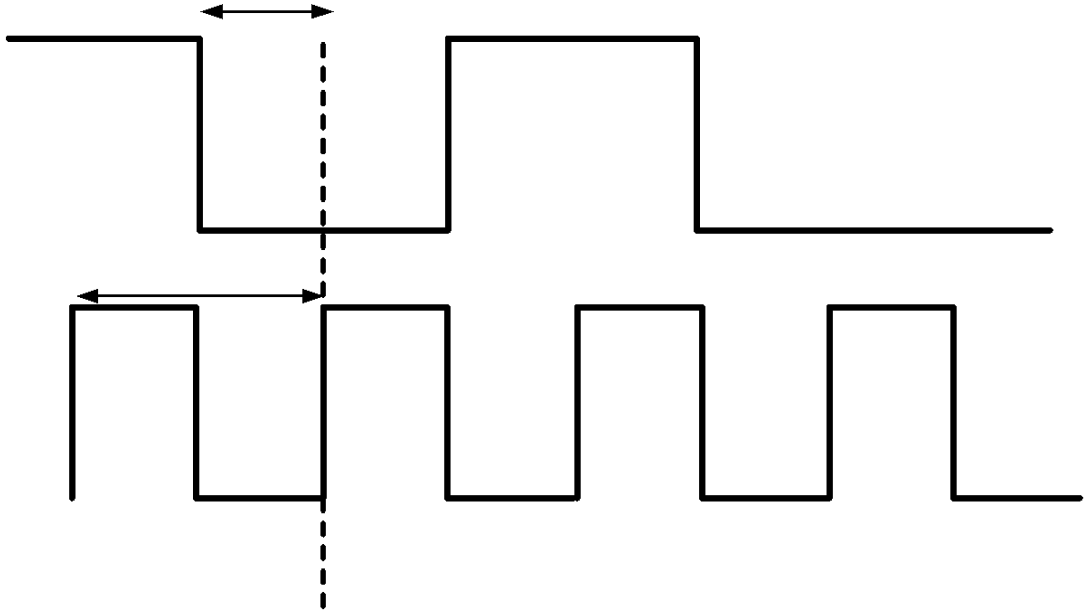


Figure 4. Timing Diagram for V3 mode.

5.3 Reset/Sync Timing

LArPix-v3 can execute three operations through assertion of the active-low RST_SYNC pad (in other words, it is kept high naturally and pulled low to assert it). The behavior of RST_SYNC is summarized in Table 4.

To enable the timestamp to be synchronized using the sync pin, the sync functionality must be enabled by the user by setting **enable_external_sync** to 1.

Number of clock cycles RST_SYNC held low	**Action**
More than 4	Sync timestamp (set to 0)
More than 15	Internal reset (reset state machines but do not reset configuration memory)

More than 32	Full reset (reset all internal state machines as well as reset configuration memory)

Table 4. RST_SYNC behavior.

6 LArPix-v3 Specifications

6.1 Power Supplies

The power supply specifications for LArPix-v3 are summarized in Table 5.

UPDATE after measurements!

Specification	Padring Name	Value	Units	Note
Analog Power Supply	VDDA	2.5 nominal	V	VDDA ? 1.9 mA @ 2.5V
Digital Logic Power Supply	VDDD	1.2 nominal (functional down to 1 V)	V	VDDD ? 5 mA @ 10 MHz System Clock, 1.2 V supply
Digital I/O Power Supply	VDDIO	2.5 nominal (functional at 88 K down to 1.5 V)	V	
Total Power Dissipation		~ 3.4 mW analog ~ 9 mW digital ---------- 12.4 mW total (~ 200 µW/channel)		150 µW / channel @ 300 K & 10 MHz CLK operation (1 Hz average hit rate) Can be significantly improved by reducing operating voltages.

Table 5. LArPix-v3 power supply specifications.

6.2 Analog Specifications

The analog specifications for LArPix-v3 are summarized in Table 6.

Specification	Value	Units	Note
Number of Analog Inputs (channels)	64 (single-ended)		
Noise	300 @ 88K 500 @ 300K	ENC	
Time resolution	2	µs	with 10 MHz clock rate
Analog Dynamic Range	~1300	mV	max signal ~ 20 MIP, minimum detectable signal ~0.04 MIP (+ 6 dB rms noise)
ADC resolution	10	bits	programmable LSB, 4 mV nominal (= 0.1 fC or 600 e-)
Threshold Range	0 – 1.8	V	
Threshold Resolution	< 1	mV	nominal
Channel Linearity	1	%	
Operating Temperature Range	80 - 300	°K	

Table 6. Analog Specifications.

6.3 Digital Specifications

The digital specifications for LArPix-v3 are summarized in Table 7.

Specification	Value	Units	Note

Event Memory Depth	2048	memory locations	~8 ms without data loss in case of track normal to pixel plane
Output Signaling Level	Programmable	V	Programmable, differentially LVDS-like signaling.
Digital data rate	5	Mb/s	with 10 MHz System Clock
Event readout time	12.8	µs	

Table 7. Digital Specifications.

7 LArPix-v3 Configuration

7.1 LArPix-v3 Channel

The high-level design for the LArPix-v3 ASIC is similar to many readout ASICs for capacitive radiation detectors with the exception that a shaping amplifier is not included in LArPix-v3. This is possible because the noise specifications of the ASIC are significantly relaxed because the detector capacitance is much less than expected for the LArASIC ASIC, for instance. Removing the shaper not only greatly reduces power dissipation it also simplifies and accelerates the design of the ASIC as many important and challenging design considerations are eliminated. A conceptual block diagram of the channel is shown in Figure 5.

The channel operates as follows. The charge input from the detector is integrated on the feedback capacitance (labeled C) of a charge-sensitive amplifier (CSA). The CSA followed by a buffer amplifier and the buffer output is fed to the input of a discriminator with a per-channel programmable threshold and to a fully differential, 10-bit, Asynchronous SAR ADC. The discriminator fires if the CSA output reaches the point where it exceeds the discriminator threshold. The digital controller then initiates a digitization of the buffer output voltage and resets the CSA so it is ready for the next charge packet.

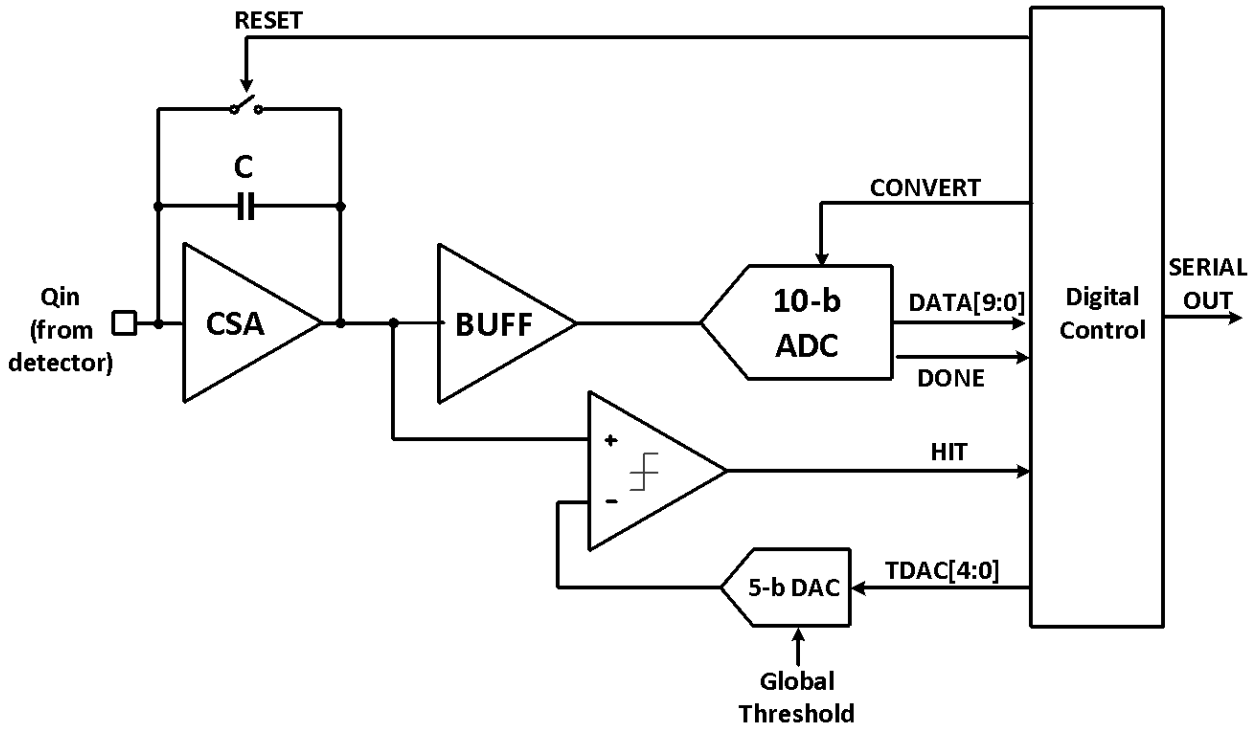


Figure 5. Conceptual block diagram of LArPix-v3 channel.

Various aspects of the data acquisition process are user-programmable as described below. The ADC LSB is determined by

$$LSB = \frac{2*V_{REF}}{2^N - 1} = \frac{2*V_{REF}}{1023}$$

The ADC output is related to its input voltage as follows:

$$D_{out} = \frac{V_{in}}{V_{REF}}(2^{N-1} - 1) + 2^{N-2}$$

V_{REF} sets the ADC maximum input and V_{CM} sets the ADC common-mode voltage, and N=10. During normal operation, V_{CM} should be set to VDD/2 for the ADC to function properly. This reflects the fact that no differential drive has been implemented yet and the negative terminal of the ADC is connected to V_{CM}.

VCM is set via **vcm_dac[7:0]** and VREF is set **vref_dac[7:0]**. In both cases the reference voltage is set to the DAC value divided by 256 times the power supply voltage.

7.2 Configuration Register Map

The configuration register map for LArPix-v3 is shown in Table 8[2].

Signal Name	Description	Reg	# of bits	Default	Comment
pixel_trim_dac	Pixel-level threshold DAC for channel X	0[4:0] – 63[4:0]	5	5'b0	Each channel has independent trim setting
threshold_global	Global threshold DAC (shared across all channels)	64	8	8'hFF	All channels disabled at power up
csa_bypass_enable	Enables bypass of CSA and injects signal from ADC_TEST directly into selected ADCs	65[1]	1	1'b0	
bypass_caps_en	Enable internal supply bypassing	65[2]	1	1'b1	Active high (default is internal bypassing enabled)
csa_enable	High to enabled CSA (otherwise leave in hard reset)	66-73	64	64'b1	Power up with all CSAs in hard reset. Ex: 73[7] = channel 63
ibias_tdac	Threshold DAC bias current	74[3:0]	4	4'b1000	LSB = IREF/30 (see below)
ibias_comp	Discriminator bias current	75[3:0]	4	4'b1000	LSB = IREF/30 (see below)
ibias_buffer	Channel Buffer bias current	76[3:0]	4	4'b1000	LSB = IREF/10

[2] The defaults are based on Verilog HDL notation. The number before the ' indicates the number of bits or hex digits in the number, the letter b or h after ' indicates whether the number is in binary or hexadecimal, and a single value after the letter indicates all bits or hex digits are identical. A single 0 or 1 means the bit is repeated. For example, 8'b0 means 0000_0000 (or eight individual bits set to zero) and 8'h03 means the 8-bit number 0000_0011.

ibias_csa	CSA bias current	77[3:0]	4	4'b1000	LSB = 4*IREF/25
ibias_vref_buffer	VREF buffer bias current	78[3:0]	4	4'b1000	LSB = IREF/15 (active low): ibias = (15-code)*LSB
ibias_vcm_buffer	VCM buffer bias current	79[3:0]	4	4'b1000	LSB = 2*IREF/15 (active low): ibias = (15-code)*LSB
ibias_tpulse	CSA pulser bias current	80[3:0]	4	4'b0101	LSB = IREF/10
ref_current_trim	Trim for IREF (primary reference current)	81[4:0]	5	5'b1_0000	IREF = 30 µA nominal (IREF = 1.5*I_READBACK). @warm: IREF = (24.5 + 0.5*code) µA @cold: IREF = (22.75 + 0.4*code) µA
vref_dac	DAC to set VREF for ADC	82	8	8'hDB	VREF = code*VDDA/256
adc_ibias_delay	Bias for ADC delay chain	83[3:0]	4	4'b1000	LSB = IREF/30 (see below) Can be used to tweak ADC speed.
adc_ibias_delay_monitor	High to enable monitor (one-hot)	83[7:4]	4	4'b0	Shorted to current_monitor (reg 109 & reg 110)
adc_comp_trim	Bias for ADC delay discriminator	??	2	2'b10	Nominally set to ~VDDA/2. Can be used to tweak ADC speed.
csa_bypass_select	Selects ADC channel(s) for test.	84[0] – 91[7]	64	64'b0	csa_bypass_enable must = 1 for this configuration to have effect. 84[0] is channel 0.
csa_monitor_select	Selects channel to connect to analog monitor bus	92[0] – 99[7]	64	64'b0	Active high (set to 1 to select channel). 92[0] is channel 0. Only select one channel at a time.

csa_testpulse_enable	Selects channel(s) to connect to CSA test pulse	100[0] – 107[7]	64	64’b1	Active low (Should be set to high in normal operation). 100[0] is channel 0.
csa_testpulse_dac[7:0]	Set test pulse amplitude	108	8	8’b0	see section on test pulse
current_monitor_bank0	Current monitor for bank 0. Current routed to ANA_MONITOR pin	109[3:0]	5	5’b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp bit[0]: ibias_tdac
current_monitor_bank1	Current monitor for bank 1. Current routed to ANA_MONITOR pin	109[7:4]	5	5’b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp bit[0]: ibias_tdac
current_monitor_bank2	Current monitor for bank 2. Current routed to ANA_MONITOR pin	110[3:0]	5	5’b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp bit[0]: ibias_tdac
current_monitor_bank3	Current monitor for bank 3. Current routed to ANA_MONITOR pin	110[7:4]	5	5’b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp bit[0]: ibias_tdac
voltage_monitor_bank0	Voltage monitor for bank 0. Voltage routed to ANA_MONITOR pin	111[2:0]	3	3’b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa

voltage_monitor_bank1	Voltage monitor for bank 1. Voltage routed to ANA_MONITOR pin	111[5:3]	3	3'b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa
voltage_monitor_bank2	Voltage monitor for bank 2. Voltage routed to ANA_MONITOR pin	112[2:0]	3	3'b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa
voltage_monitor_bank3	Voltage monitor for bank 3. Voltage routed to ANA_MONITOR pin	112[5:3]	3	3'b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa
voltage_monitor_refgen	Voltage monitor for reference generation block. Voltage routed to ANA_MONITOR pin. The "set" voltages are raw DAC outputs (before buffering, used to verify DAC functionality)	113[7:0]	8	8'b0	bit[7]: vref_adc bit[6]: vref_adc_set bit[5]: vcm_adc bit[4]: vcm_adc_set bit[3]: csa_tpulse_internal bit[2]: csa_tpulse bit[1]: csa_thresh_global bit[0]: biasp1
digital_monitor_enable	High to enable digital monitor port	114[0]	1	1'b0	
digital_monitor_select	Selects digital monitor output signal	114[4:1]	4	4'b0	10: sync_timestamp 9: reset_n_config_syn 8: cross_trigger 7: external_trigger 6: periodic_reset

					5: periodic_trigger 4: triggered_natural 3: csa_reset 2: sample 1: comp 0: hit
digital_monitor_chan	Selects the channel to observe with monitor	115[5:0]	6	6'b0	
fifo_high_water_lsbs	LSBs of digital measurement of highest FIFO occupancy since last reset	116	8	8'b0	
fifo_high_water_msbs	3 MSBs of digital measurement of highest FIFO occupancy since last reset	117	3	3'b0	
total_packets_lsbs	LSBs of total number of packets generated by current chip in FIFO since last system reset	118	8	8'b0	
total_packets_msbs	MSBs of total number of packets generated by current chip in FIFO since last system reset	119	8	8'b0	
dropped_packets	Total number of rejected packets (bad ID or magic number) since last reset	120	8	8'b0	

adc_hold_delay	Number of clock cycles for sampling in ADC (i.e. how many cycles to wait before transitioning the SAR into hold mode	120[7:0] -121[7:0]	16	16’0	In normal operation, the SAR is in sampling mode for at least three clock cycles (track/hold operation). ADC hold delay adds additional clock cycles for sampling at the expense of dead time. High order bits in register 121.
chip_id	Unique ID used to address and identify chip	122	8	8’h01	Default ID is 1. Broadcast ID is 255 (all chips respond to commands with Chip ID = 255, regardless of their individual IDs).
cds_mode	Correlated Double-Sampling Mode. High to configure ADC to provide pedestal sample for each packet (or packet burst).	123[0]	1	1’b0	0 → CDS disabled 1 → CDS enabled
load_config_defaults	Active high.	123[1]	1	1’b0	Must be set low again immediately to allow further interaction with registers.
enable_FIFO_diagnostics	High to embed FIFO counter information in output word. Replaces bits [15:12] of timestamp	123[2]	1	1’b0	When active, bits [43:32] are 12-bit shared FIFO counter value at time event is latched, bits [45:44] are 2-bit local FIFO counter, and bits [47:46] are 0.
enable_local_fifo_diagnostics	High to embed local FIFO usage in output packets	123[3]	1	1’b0	0 → Do not embed local FIFO information in data packet

					1 → Embed local FIFO information in data packet
enable_tally	High to embed 4 LSBs of total_packets in output packets	123[4]	1	1'b1	0 → local FIFO half/full 1 → 4 LSBs of tally
enable_external_trigger	High to enable external trigger	123[5]	1	1'b0	0 → disable external trigger 1 → enable external trigger
enable_external_sync	High to enable timestamp to be synced by external signal	123[6]	1	1'b0	0 → disable external sync 1 → enable external sync
enable_data_stats	High to record packet data (high water, dropped packets, total packets) in configuration registers	123[7]	1	1'b1	0 → disable data stats 1 → enable data states
enable_PISO_upstream	Enable different PISOs for upstream operation. (For data originating in FPGA)	124[3:0]	4	4'b0	bit[3]: MIS03 bit[2]: PISO2 bit[1]: PISO1 bit[0]: PISO0
enable_PISO_downstream	Enable different PISOs for downstream operation (for data destined for FPGA)	125[3:0]	4	4'b0	bit[3]: MIS03 bit[2]: PISO2 bit[1]: PISO1 bit[0]: PISO0
enable_POSI	Enable different POSIs	126[3:0]	4	4'b1	bit[3]: M0SI3 bit[2]: POSI2

					bit[1]: POSI1 bit[0]: POSI0
en_analog_monitor	Powers up and connects analog monitor	127[0]	1	1’b0	The analog monitor uses significant power so disable it when not in active use.
enable_cross_trigger	Enable LArPix-v3 cross trigger	128[0]	1	1’b0	All unmasked channels read out when any channel hit
enable_periodic_reset	Channels reset periodically even if no hit. All channels reset simultaneously.	128[1]	1	1’b0	Number of clocks between resets specified in periodic_reset_cycles (registers 163-165)
enable_rolling_periodic_reset	Channel is reset periodically even if no hit. Channels reset in a rolling fashion (one at a time)	128[2]	1	1’b0	Number of clocks between resets specified in periodic_reset_cycles (registers 163-165)
enable_periodic_trigger	Channels trigger periodically even if no hit. All unmasked channels trigger simultaneously.	128[3]	1	1’b0	Number of clocks between triggers specified in periodic_trigger_cycles (registers 166-169)
enable_rolling_periodic_trigger	Channel is triggered periodically even if no hit. Channels triggered in a rolling fashion (one at a time)	128[4]	1	1’b0	Number of clocks between triggers specified in periodic_reset_cycles (registers 163-165)
enable_periodic_triger_veto	High to enable the hit signal to veto the periodic trigger	128[5]	1	1’b1	If high, periodic trigger is skipped if channel servicing a hit
enable_hit_veto	High to enforce that discriminator hit is high before channel	128[6]	1	1’b1	If high, channel is not triggered on transient discriminator hits

	goes into ADC hold mode				
shadow_reset_length	Number of clock cycles to hold CSA reset when asserted	129	8	8'b0	OR'd with reset_length (register 170[5:3])
adc_burst_length	Number of ADC conversions per channel hit	130	8	8'b0	Maximum burst length = 255 ADC conversions
channel_mask	High to disable channel	131-138	64	64'b1	131[0] = channel 0
external_trigger_mask	High to disable external trigger for a channel	139-146	64	64'b1	
cross_trigger_mask	High to disable cross trigger for a channel	147-154	64	64'b1	
periodic_trigger_mask	High to disable periodic trigger for a channel	155-162	64	64'b1	
periodic_reset_cycles	Number of clock cycles between periodic resets	163-165	24	24'h00_10_00	
periodic_trigger_cycles	Number of clock cycles between periodic triggers	166-169	32	32'h0	
enable_dynamic_reset	High to enable resetting channel based on ADC data.	170[0]	1	1'b0	Set **adc_burst_length** to maximum desired conversion
enable_min_delta_adc	High to enable mode that resets channel when ADC values have settled	170[1]	1	1'b0	**enable_dynamic_reset** must be low for this bit to have effect

threshold_polarity	High to trigger when ADC above threshold, low to trigger when ADC below threshold	170[2]	1	1’b1	
reset_length	Number of clock cycles to hold CSA reset when asserted	170[5:3]	3	3’b001	Setting this to 0 has same effect as setting to 1
mark_first_packet	Sets MSB of timestamp to '1' for first packet, '0' for other packets	170[6]	1	1’b1	
reset_threshold	In dynamic mode, only reset when ADC is larger than this value	171	8	8’b0	
min_delta_adc	A difference in successive conversions less that this value triggers a channel reset (and ends the ADC burst). This word only has effect if enable_dynamic_reset is high and enable_min_delta_adc is low). This is modulated by threshold polarity.	172	8	8’b0	For example, if threshold polarity == 1, and min_delta_adc == 5, then adc[x] = 200 and adc[x-1] = 197 would trigger a reset, but adc[x] = 200 and adc[x-1] = 203 would not.
digital_threshold	Only put words into FIFO if they are at least this value	173-236	8	8’b0	Word 173 corresponds to channel 0
RESERVED		237-238			Unused in LArPix-v3

tx_slices0	Number of slices to enable for PISO0 link	239[3:0]	4	4'h8	Active high. Setting tx_slices0 = 4'b0 disables PISO0.
tx_slices1	Number of slices to enable for PISO1 link	239[7:5]	4	4'h8	Active high. Setting tx_slices1 = 4'b0 disables PISO1.
tx_slices2	Number of slices to enable for PISO2 link	240[3:0]	4	4'h8	Active high. Setting tx_slices2 = 4'b0 disables PISO2.
tx_slices3	Number of slices to enable for PISO3 link	240[7:5]	4	4'h8	Active high. Setting tx_slices3 = 4'b0 disables PISO3.
i_tx_diff0	Differential per-slice loop current for PISO0 link	241[3:0]	4	4'hC	Active low. LSB = IREF/30.
i_tx_diff1	Differential per-slice loop current for PISO1 link	241[7:4]	4	4'hC	Active low. LSB = IREF/30.
i_tx_diff2	Differential per-slice loop current for PISO2 link	242[3:0]	4	4'hC	Active low. LSB = IREF/30.
i_tx_diff3	Differential per-slice loop current for PISO3 link	242[7:4]	4	4'hC	Active low. LSB = IREF/30.
i_rx0	Receiver bias current for POSI0 link	243[3:2]	2	2'b10	Active Low. LSB = IREF/50. 2 LSBs hardwired to 2'b00 (see text).
i_rx1	Receiver bias current for POSI1 link	243[7:6]	2	2'b10	Active Low. LSB = IREF/50. 2 LSBs hardwired to 2'b00 (see text).
i_rx2	Receiver bias current for POSI2 link	244[3:2]	2	2'b10	Active Low. LSB = IREF/50. 2 LSBs hardwired to 2'b00 (see text).

i_rx3	Receiver bias current for POSI3 link	244[7:6]	2	2'b10	Active Low. LSB = IREF Active Low. LSB = IREF/50. 2 LSBs hardwired to 2'b00 (see text).
i_rx_clk	Receiver bias current for clock rx	245[3:2]	2	2'b10	Active Low. LSB = IREF/50. 2 LSBs hardwired to 2'b00 (see text).
i_rx_rst	Receiver bias current for reset rx	245[7:6]	2	2'b10	Active Low. LSB = IREF/50. 2 LSBs hardwired to 2'b00 (see text).
i_rx_ext_trig	Receiver bias current for external trigger rx	246[3:2]	2	2'b10	Active Low. LSB = IREF/50. 2 LSBs hardwired to 2'b00 (see text).
r_term0	Termination resistor for POSI0 link	247[2:0]	3	3'h4	Total resistance = 2.5 kΩ in parallel (r_term) times (see text). Hi-Z when r_term1 = 0. Default is 1.25 kΩ. (2 MSBs hardwired to 1'b0 as described in text)
r_term1	Termination resistor for POSI1 link	248[2:0]	3	3'h2	Total resistance = 2.5 kΩ in parallel (r_term) times (see text). Hi-Z when r_term1 = 0. Default is 1.25 kΩ. (2 MSBs hardwired to 1'b0 as described in text)
r_term2	Termination resistor for POSI2 link	249[2:0]	3	3'h2	Total resistance = 2.5 kΩ in parallel (r_term) times (see text). Hi-Z when r_term1 = 0. Default is 1.25 kΩ. (2 MSBs hardwired to 1'b0 as described in text)
r_term3	Termination resistor for POSI3 link	250[2:0]	3	3'h2	Total resistance = 2.5 kΩ in parallel (r_term) times (see text). Hi-Z when r_term1 = 0. Default is 1.25 kΩ. (2 MSBs

					hardwired to 1'b0 as described in text)
v_cm_lvds_tx0	Common-mode voltage for PISO0 link	254[2:0]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.
v_cm_lvds_tx1	Common-mode voltage for PISO1 link	254[6:4]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.
v_cm_lvds_tx2	Common-mode voltage for PISO2 link	255[2:0]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.
v_cm_lvds_tx3	Common-mode voltage for PISO3 link	255[6:4]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.

Table 8. LArPix-v3 configuration register map.

7.3 Channel Threshold

The channel threshold consists of two components. A global threshold set by **threshold_global[7:0]** and a per-channel trimmable threshold set by **pixel_trim_dac[4:0]**.

The global threshold value is:

$$Th_{Gl} = \frac{VDDA}{256} threshold_global[7:0]$$

At room temperature (300K) each channel threshold is:

$$Th_chan_RT_X = Th_{Gl} + 210\,mV + 1.45mV * pixel_trim_dacX[4:0]$$

At 88 K each channel threshold is:

$$Th_chan_88K_X = Th_{Gl} + 465\,mV + 2.34mV * pixel_trim_dacX[4:0]$$

Please keep in mind the absolute accuracy of the thresholds is greatly affected by mismatch (especially at cold).

8 Digital I/O

LArPix-v3 includes a programmable current mode digital I/O to reduce crosstalk. A block diagram of the system is shown in Figure 6. The transmitted bit is converted to a differential current at the transmitter chip and sent to the receiver chip. At the receiver chip, the differential current is converted to a differential voltage by a termination resistor (which may be internal or external to the receiving chip). In the receiving chip, an amplifier senses the differential voltage and outputs a logic-level CMOS bit.

The POSI RX circuits are terminated internally using a digitally programmable resistor. The CLK, RST/SYNC and External Trigger inputs are shared at the tile level and are terminated on the tile into 100 Ω differential.

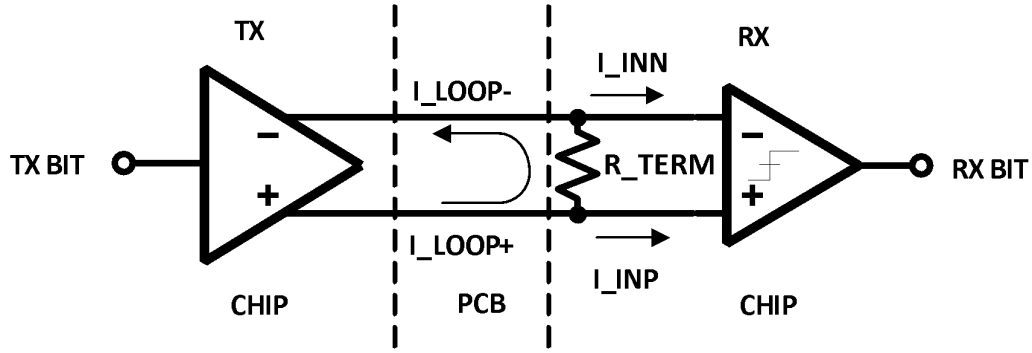


Figure 6. LArPix-v3 differential digital I/O.

The common-mode voltage (the average voltage of the two parts of the loop) and the loop current can be adjusted on the TX side and the termination resistance and receiver bias current can be adjusted on the RX side. The loop current can be adjusted in two ways. The transmitter is implemented by 15 parallel transmit modules. For large changes in loop current, additional transmit slices can be added or removed from operation. For fine tuning, the raw per-slice current can also be adjusted. This ensures proper biasing at cold even over a wide range of possible loop currents.

8.1 Digital I/O TX Configuration

The Digital I/O transmitter can be configured by adjusting the number of parallel slices engaged, the loop current contributed by each slice, and the common-mode voltage of the transmission.

There are 15 parallel slices. The number of slices connected to the output can be adjusted by changing **tx_slicesX[3:0]** (where, for instance, **tx_slices0[3:0]** controls the number of slices contributing to the PISO0 loop current.

The amount of current contributed by each slice is controlled by the active low control **i_tx_diffX[3:0]**.

The loop current in the link is defined by the following equation:

i_loop_tx = **tx_slicesX[3:0]** * (15- **i_tx_diffx[3:0]**)*i_ref/10. Therefore, the maximum current (when i_ref is adjusted to its nominal value of 30 µA) is 675 µA

i_ref is the Primary bias current that is set by the on-chip biasing module (see below for information on adjusting i_ref).

The common-mode voltage of the transmitter is set by adjusting the active high control **v_cm_lvds_txX[2:0**]. The LSB of the DAC that sets of common-mode voltage is VDD/8.

8.2 Digital I/O RX Configuration

The Digital I/O receiver can be configured by adjusting its bias current and its internal termination resistor.

The bias current in the receiver is adjusted using the active low control **i_rxX[3:0]**. The value of the bias current is:

i_bias_rx = (15 – **i_rxX[3:0]**)* i_ref/50.

i_ref is the primary current set but the on-chip biasing module (see below for information on adjusting i_ref).

The termination resistor is set by adjusting **r_termX[4:0]**. The termination resistor is implemented using internal 2.5 kΩ resistors. So, the termination resistor is approximately 2.5 kΩ/**r_termX[4:0]**. When **r_termX[4:0]** is set to zero, then the receiver is high impedance and the channel must be terminated external to LArPix-v3.

8.3 Digital I/O Configuration Summary

The settings to adjust the various controls available for the Digital I/O are summarized in Table 9. The "X" in each control name indicates which PISO/POSI port it is affecting.

Configuration Word Name	Adjustment	Bias Transfer Function LSB
tx_slicesX[3:0]	Number of parallel TX slices engaged	LSB = unit slice
i_tx_diffX[3:0]	Loop current contributed per slice	LSB = i_ref/30 (active low)
v_cm_lvds_txX[2:0]	Transmitter common-mode output voltage	LSD = VDD/8
i_rxX[3:0]	Bias current for receiving amplifier	LSB = i_ref/50 (active low)
r_termX[2:0]	Input termination resistor for receiving amplifier	LSB = 2.5 kΩ resistor. Word indicates number of 2.5 kΩ unit devices that comprise termination resistor.

Table 9. Digital I/O Configuration Summary.

9 Hydra Network

9.1 Introduction to Hydra

The Hydra I/O is the organizational architecture for connecting multiple LArPix-v3 chips into a network. Because of the large number of LArPix-v3 chips used in a practical system, it will be important to limit the physical lines of communications (both to reduce the number of cryostat penetrations and to simplify PCB design). The approaches used in LArPix-v1 and LArPix-v3 are contrasted in Figure 7. LArPix-v1 had a single UART per chip and used a simple daisy chaining approach to satisfy these requirements. Each LArPix-v1 ASIC was connected to its two nearest neighbors in one dimension and all the ASICs were organized in a series of columns. All data would be forward to the next downstream ASIC until it would finally exit the cryostat. As is, this approach is simple and easy to implement but it not scalable since a

single ASIC failure would make it impossible to communicate with all ASICs upstream of the failed ASIC on the daisy chain.

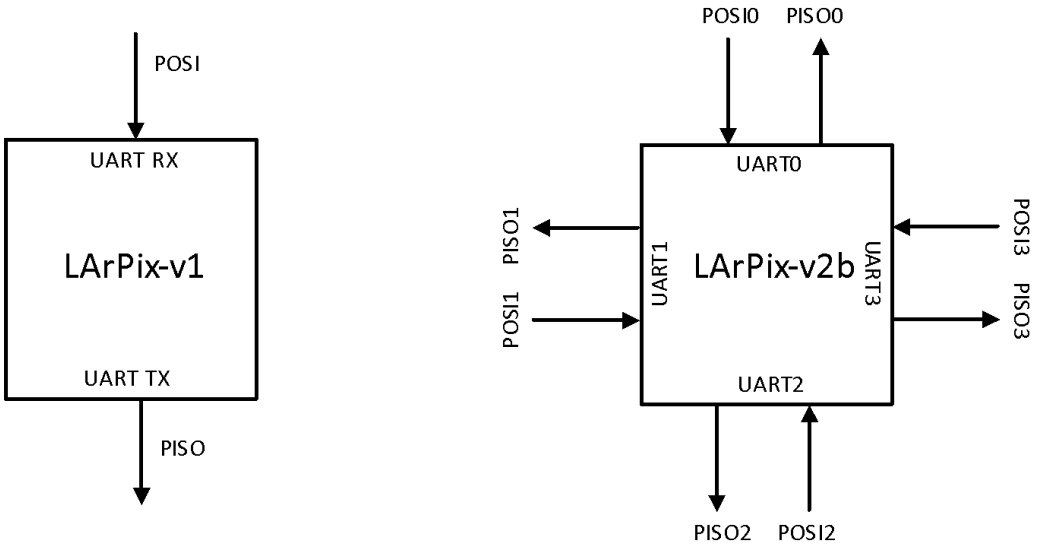


Figure 7. Comparison of available UARTs between LArPix-v1 and LArPix-v3 (figure adapted from original by Dan Dwyer).

To circumvent the reliability problems of a daisy chain approach to LArPix interconnection a new architecture called Hydra is employed. As seen in Figure 7, LArPix-v3 has four on-board UARTs and each LArPix-v3 is connected to its nearest neighbors on all four sides, and which side is the dominant direction to shift data toward the outside is configurable. Therefore, the organization is not constrained to be daisy chains and 2-D networks of arbitrary complexity can be constructed in an ad hoc way. This means single failed ASICs can be easily routed around (which was problematic from an implementation standpoint in the daisy chain approach).

In Hydra there are downstream and upstream channels. Upstream channels are used to send configuration commands through the network. Downstream channels are used to return data and requested configuration data to the controller. Downstream and upstream channels use the same PISO ports so the difference is logical rather than physical. Downstream packets are marked as such to differentiate them from upstream packets.

While Hydra provides the means by which large numbers of LArPix-v3 chips and collaborate, each individual LArPix-v3 ASIC knows nothing about Hydra I/O and is only be able to power up and down UARTs and respond to commands that match its Chip ID. It will be up to the configuration script author to ensure that pathological things (like configuration infinite loops) do not occur. Caveat Emptor.

9.2 Chip ID

Each individual ASIC in a Hydra network requires an individual ID that allows it to be addressed. LArPix ASICs will only respond to commands that are addressed to their chip ID or to the broadcast chip ID of 255. All commands sent to Chip ID 255 will be processed by all ASICs. To change the chip ID of an individual LArPix ASIC, update the value in the register **chip_id[7:0]**.

9.3 Example Hydra Network

An example Hydra network of 25 LArPix-v3 ASICs arranged in a 5-by-5 grid is shown in Figure 8.

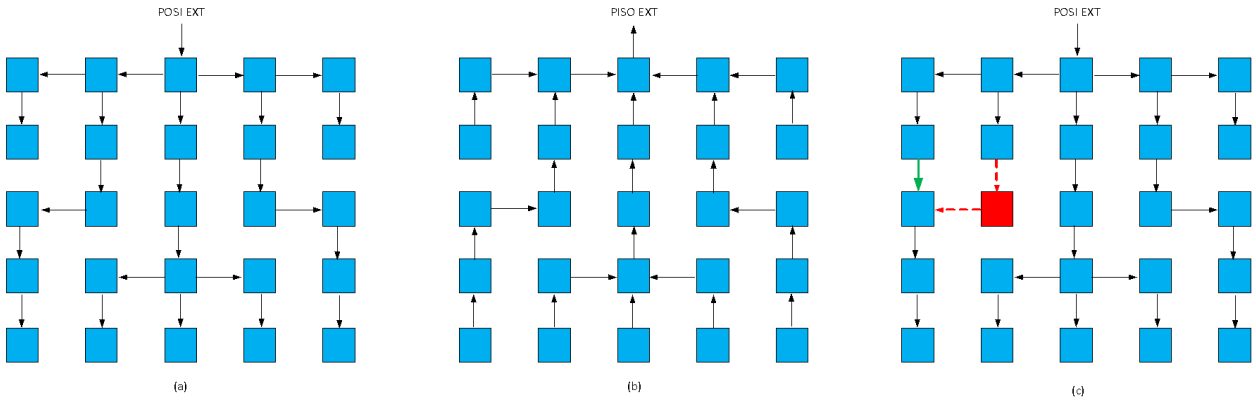


Figure 8. Hydra Network. a) upstream network, b) downstream network, c) upstream network re-routing around damage. (figure adapted from original by Dan Dwyer)

The upstream Hydra network in Figure 8a is used to configure the various LArPix-v3 ASICs in the network. Each chip is given a unique Chip ID and POSI ports are enabled to make an efficient topology to configure a large number of ASICs. The corresponding upstream network is shown in Figure 8b. In the case of an ASIC failure, the Hydra network can be reconfigured to route around the damage. This is shown in Figure 8c. The red ASIC has failed and the red, dotted connections are no longer operable. Without Hydra's ability to reconfigure, not only the red chip would be lost but all downstream ASICs as well would be out of communication. Using Hydra, however, the green connection can be configured by enabling a different POSI port and all of the chips in the network are once again reachable.

9.4 UART Configuration

Each of the four UARTs on LArPix-v3 can be configured independently. Each POSI can be enabled or disabled (although care should be taken to make sure the ASIC can still be addressed) using the configuration bits **enable_POSI[3:0]**, where each configuration bit corresponds to a single POSI output. The PISO ports can be configured as upstream or downstream using the **enable_PISO_upstream[3:0]** and **enable_PISO_downstream[3:0]** bits.

9.5 Hydra Network Bringup

To bring up a Hydra network, during configuration each ASIC will need to:

- Update Chip ID to a unique value (default Chip ID is 1)
- Update configuration registers as desired
- Enable downstream PISO to all configuration read commands
- Verify all registers are correctly configured
- Enable appropriate PISO channel for one desired upstream channel
- Contact next upstream ASIC and configure
- Repeat for each ASIC in desired Hydra network configuration

The basic idea is that only one chip that is actively attached to the network will have Chip ID at 1 at any time. Otherwise, it may become ambiguous which chip is currently under configuration.

As an example, consider the simple four-chip Hydra network in Figure 9. There is a single connection to the outside work (FPGA) that will be used to configure the entire network. At power up, each chip has Chip ID = 1. To configure the network as shown the following commands can be issued from the controlling FPGA (or computer) as seen in Table 10.

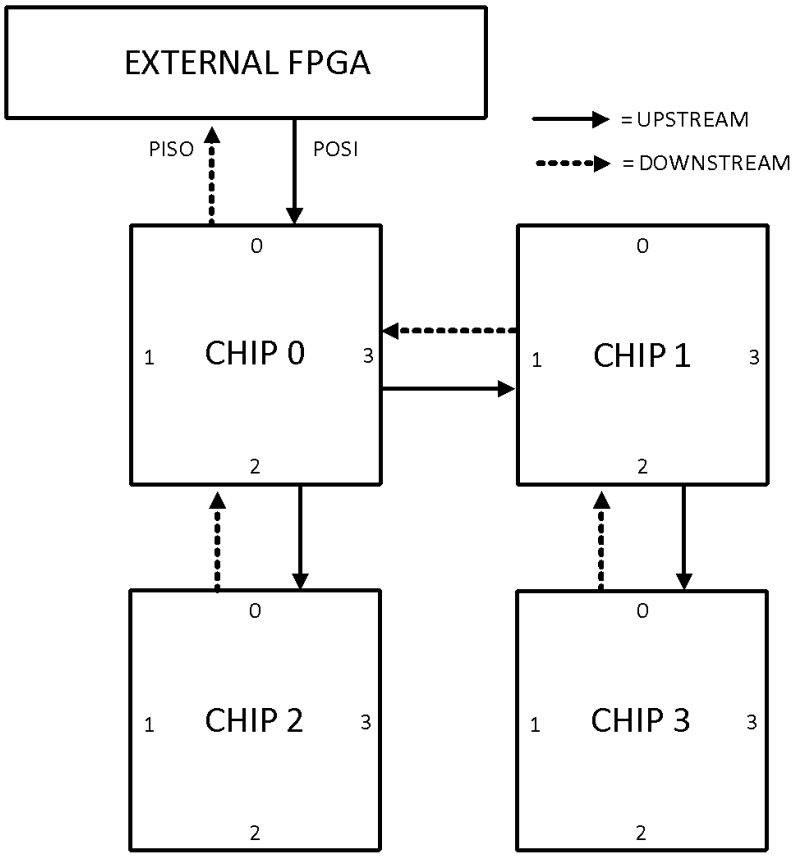


Figure 9. Four-chip Hydra network.

Step	Chip ID	Config Signal Name	Value	Explanation
1	1	chip_id	4	Change Chip ID of Chip 0 from 1 to 4.
2	4	enable_PISO_downstream	1	Enable Chip 0 - PISO0 to return downstream data to FPGA
3	4	enable_PISO_upstream	8	Enable Chip 0 – PISO3 (for writing data to chip 1)
4	1	chip_id	2	Change Chip ID of Chip 1 from 1 to 2.
5	2	enable_PISO_downstream	2	Enable Chip 1 – PISO1 to return downstream data to Chip 0

6	2	enable_PISO_upstream	4	Enable Chip 1 – PISO2 (for writing data to chip 3)
7	1	chip_id	3	Change Chip ID of Chip 3 from 1 to 3.
8	3	enable_PISO_downstream	1	Enable Chip 3 – PISO0 to return downstream data to Chip 1
9	4	enable_PISO_upstream	4	Enable Chip 0 – PISO3 (for writing data to chip 2)
10	1	chip_id	5	Change Chip ID of Chip 3 from 1 to 5.
11	5	enable_PISO_downstream	1	Enable Chip 2 – PISO0 to return downstream data to Chip 0

Table 10. Algorithm to configure example Hydra network.

10 Reset Modes

LArPix-v3 has three reset modes: normal (with user defined reset stretching), periodic, and rolling periodic, and minimum delta ADC. The number of clock cycles the CSA spends in reset is determined by **reset_length[2:0]**. Setting this to 0 or 1 has the same effect (reset is asserted for one clock cycle). Increasing **reset_length[2:0]** has the benefit of making charge removal from the CSA feedback capacitor more ideal.

10.1 Normal Reset Mode

In normal reset mode, the channel is reset as soon as the sampling process has completed.

10.2 Periodic Reset

There may be situations when it is desirable to periodically reset one or more channels. For example, background events will integrate on the front-end output capacitor and will eventually cause a false hit[3]. To mitigate this, the channels can be reset periodically (at the cost of a small about of dead time). To put LArPix-v3 into periodic reset mode, set **enable_periodic_reset** to one. LArPix-v3 contains an internal counter that will trigger a reset after a given number of system clock periods. The internal counter increments every two system clocks. The number of counter increments is given by the 24 bit word **periodic_reset_cycles[23:0]** in registers 163-165. Register 165 holds the MSB byte and register 163 holds the LSB byte. The counter is incremented every four cycles of the primary clock.

Channels can be masked from responding to a periodic reset by setting **periodic_reset_mask[63:0]** to 1 for the desired channels. Note that channels automatically ignore periodic reset commands if they are currently processing a hit.

10.3 Rolling Periodic Reset

By default, all unmasked channels are reset at the same time when the reset counter equals the value in **periodic_reset_cycles[23:0]**. A separate rolling periodic reset mode is accessible by setting **enable_rolling_periodic_trigger** = 1. In this mode, only one channel is reset each time the reset counter equals the value in **periodic_reset_cycles[23:0]**.

10.4 Dynamic Reset

In dynamic reset mode, the channel does not reset until the ADC value is above (or below) the threshold specified in **reset_threshold[9:0]**. Set **threshold_polarity** = 1 for the channel to reset when the current ADC value is above the threshold and set **threshold_polarity** to 0 for the channel to reset when the ADC value is below the threshold.

10.5 Minimum Delta ADC

[3] The false hit rate is expected to be much larger at RT than at LAr temperature because of the increased device leakage (that will be integrated onto the feedback capacitor) at RT.

In this mode, the channel does not reset until the ADC values have settled to a user-specified amount. TO enable this mode, set **enable_min_delta_adc** to 1. Then, the channel will not reset itself until the difference between two consecutive ADC values is less than the value in **min_delta_adc[7:0]**. To make sure the channel does not hang, the maximum number of conversions the channel will spend waiting is set my **adc_burst_length[7:0]**. Note that **enable_dynamic_reset** must be set to 0 for this mode to function.

11 Trigger Modes

LArPix-v3 has a variety of trigger modes to allow increased flexibility during ASIC and system evaluation. Each of the trigger modes has a corresponding mask that is described below.

11.1 Hit Veto

There is a hit veto function which can be important to set correctly if various trigger modes are desired. In LArPix-v2b, the configuration bit **enable_hit_veto** = 1 required that the discriminator output is high before the channel will go into hold mode and begin the digitization process. This can be useful as a noise filter in Normal Trigger mode but will disable other trigger modes. During testing, this was considered undesirable behavior. To remedy this, the evaluation of the trigger mode and ADC Burst status happens before the hit veto is evaluated. This means that if the LArPix-v3 is in External Trigger Mode or if ADC Burst is enabled, then **enable_hit_veto** does not require the discriminator output to be high to begin the digitization process.

11.2 Normal Trigger

In normal mode, a LArPix-v3 channel triggers when the input signal exceeds the threshold of the discriminator.

11.3 Cross-Trigger

In cross-trigger mode all un-masked channels (see section on masking below) of LArPix-v3 are read out when any channel receives a hit. To put LArPix-v3 in cross-trigger mode, set **enable_cross_trigger** to 1. Be advised that in the presence of significant channel occupancy cross-trigger mode could lead to FIFO overflow.

11.4 External Trigger

LArPix-v3 can also trigger on an external signal applied to the EXT_TRIG_SYNC pin. In this case, all channels configured to respond to the external trigger signal (i.e. unmasked) will trigger when EXT_TRIG_SYNC is asserted, regardless of the input level. The trigger can be applied asynchronously and is re-synchronized internally.

To use the external trigger, it must be enabled by setting **enable_external_trigger** to 1. The EXT_TRIG_SYNC pin must be configured to accept trigger inputs.

11.5 Periodic Trigger

LArPix-v3 can trigger passed on a periodic trigger in a manner analogous to the Periodic Reset. To enable the periodic trigger functionality, assert **enable_periodic_trigger** and be sure the channels of interest are configured to respond. There is also a dedicated veto logic for the periodic trigger which can be configured by setting **enable_periodic_trigger_veto** to 1. When asserted, the otherwise properly configured channel will not respond to a periodic trigger when it is processing a natural hit.

The number of cycles between periodic triggers is set by **periodic_trigger_cycles[63:0]**.

11.6 Rolling Periodic Trigger

Similar to the rolling periodic reset, if **enable_rolling_periodic_trigger** is set to 1 then only one channel will trigger each time the periodic trigger is asserted.

11.7 ADC Burst

In some situations, it may be advantageous to take multiple ADC measurements before resetting the channel. For example, it may be more desirable to digitize the waveform as charge integrates in the analog front end rather than taking a snapshot and then resetting. Set **adc_burst_length[7:0]** to the number of additional conversions done per channel hit. After **adc_burst_length[7:0]** conversions, the channel will be reset. Setting **adc_burst_length[7:0]** to either 0x0 or 0x1 will give one conversion. ADC Burst is in effect for all trigger modes and sets a ceiling on the number of ADC conversions generated by a hit.

12 Masks

LArPix-v3 contains various masks configure which channels will respond to various trigger and reset modes.

12.1 Channel Mask

To mask off hot channels (or inhibit the response of select channels for other reasons) set the bits of **channel_mask[63:0]** corresponding to desired masked channels to one. For example, 0xFF000000F will mask out channels 24 through 31 and 0 through 3. Any hits on the masked channels will be ignored.

Important: The channel mask takes priority over all other masks. If a channel is masked using the channel mask it will not respond to any stimulus.

12.2 External Trigger Mask

It may be desirable to inhibit certain channels from responding to an external trigger. To do so, set the bits of **external_trigger_mask[63:0]** corresponding to desired masked channels to one. For example, 0xF will inhibit channels 0 through 3 from responding to an external trigger. All other channels will initiate an ADC conversion upon receiving an external trigger.

12.3 Cross Trigger Mask

It may be desirable to inhibit certain channels from responding to a cross trigger. To do so, set the bits of **cross_trigger_mask[63:0]** corresponding to desired masked channels to one. For example, 0xF will inhibit channels 0 through 3 from responding to a cross trigger. All other channels will initiate an ADC conversion upon receiving a cross trigger.

12.4 Periodic Trigger Mask

It may be desirable to inhibit certain channels from responding to a periodic trigger. To do so, set the bits of **periodic_trigger_mask[63:0]** corresponding to desired masked channels to one. For example, 0xF will inhibit channels 0 through 3 from responding to a periodic trigger. All other channels will initiate an ADC conversion upon receiving a periodic trigger.

13 Correlated Double-Sampling

LArPix-v3 provides a digital correlated double-sampling (CDS) feature to suppress pedestal noise. When CDS is enabled by setting **cds_mode** to 1, then LArPix-v3 takes an ADC sample of the CSA output directly after each reset. In the case of an event, LArPix-v3 creates an additional packet containing the pedestal (reset) sample. In the case of ADC burst mode, only one pedestal sample will be created per burst.

The use of CDS has the clear advantage in that it can effectively reject pedestal noise. However, because the two samples of the CSA is uncorrelated, using CDS increases the effective thermal noise of the ADC by a factor of $\sqrt{2}$. Therefore, CDS could be beneficial in situations where pedestal noise dominates the channel noise, but should not be used in situations where the noise is dominated by thermal noise.

In addition, the use of CDS also increases the packet volume (since a reset sample packet is included in each event)

14 FIFO usage statistics

LArPix-v3 includes the capability to store some FIFO usage statistics in the configuration register, where they can be read out using the normal configuration interface. FIFO usage statistics can be enabled by setting **enable_fifo_stats** to 1.

The available statistics are summarized in Table 11.

Configuration Name	Description	Register	Bits
fifo_high_water_lsbs	LSBs of 11-bit word containing peak FIFO usage since last reset.	116	[7:0]
fifo_high_water_msbs	MSBs of 11-bit word containing peak FIFO usage since last reset.	117	[2:0]
total_packets_lsbs	LSBs of 16-bit word containing total packets written to FIFO since last reset.	118	[7:0]
total_packets_msbs	mSBs of 16-bit word containing total packets written to FIFO since last reset.	119	[7:0]
dropped_packets	Total number of packets with incorrect chip ID or magic number rejected by LArPix_v3 since last reset.	120	[7:0]

Table 11. Available FIFO usage statistics.

15 Digital Threshold

Depending on operational conditions, it may be known that ADC values below a certain value are due to noise hits and are not valid data. Therefore, LArPix-v3 has a digital threshold functionality that inhibits on a per-channel basis the placing of events in the FIFO if the ADC is below the digital threshold. This has the positive impact of reducing power and pressure on the FIFO. The 10-bit **digital_threshold[9:0]** word is broken into a shared 5-bit MSB word and a per-channel 5-bit LSB word. The shared MSBs digital_threshold[9:7] are stored in Register 173 and 174. **digital_threshold[9:7]** is stored in Register 173[7:5] and **digital_threshold[6:5]** is stored in Register 174[6:5]. The LSBs for the per-channel **digital_threshold[4:0]** are stored in Registers 173 - 236. Register 173[4:0] corresponds to Channel 0 and Reg 236[4:0] corresponds to Channel 63.

16 Monitors

LArPix-v3 has two monitor busses, one analog and one digital. This enables various internal signals to be observed on the bench during evaluation of the prototype. The CSA monitor and Analog monitor use the same physical bus so care must be taken not to short them out (this is not enforced in hardware).

16.1 CSA Monitor

The analog output of each channel can be monitored using the analog monitor bus (described in the next section. The command is active high, so to connect a channel to the analog monitor bus, set (i.e. set to high) the corresponding bit of **csa_monitor_select[63:0]**. For example, to connect the output of channel 7's analog front end to the analog monitor bus, set **csa_monitor_select[63:0]** to 0x8. Only one channel at a time should be connected to the analog monitor bus across all chips connected to a single bus. This is not enforced in hardware so care should be taken to avoid shorts made by connecting multiple channels to the analog monitor bus.

16.2 Analog Monitor

The analog monitor is used to view various internal signals through the ADC_TEST pins. The analog monitor is a differential output driver that can be used to monitor various bias voltages as well as the CSA output voltages. When **en_analog_monitor** is high, the ADC_TEST pads are connected to the output of the analog monitor. When **en_analog_monitor** is low, the ADC_TEST pads are connected to the ADC inputs.

The biasing is split into quadrants in the chip and therefore the bias voltages of the different quadrants can be monitored to see if they are consistent across the chip. The analog monitor bits are all one-hot so setting any of them to 1 will make the connection. Therefore, care must be taken to avoid shorting the bus (it is not enforced in hardware). The signals available for monitoring are summarized in Table 12.

Configuration name	Bank Monitored	Signal Monitored	Voltage / Current
voltage_monitor_bank0[2:0]	0	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage

voltage_monitor_bank1[2:0]	1	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage
voltage_monitor_bank2[2:0]	2	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage
voltage_monitor_bank3[2:0]	3	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage
voltage_monitor_refgen[2:0]	N/A	bit[0] --> biasp1 bit[1] --> csa_glob_thresh bit[2] --> csa_tpulse bit[3] --> csa_tpulse_int bit[4] --> vcm_adc_set bit[5] --> vcm_adc bit[6] --> vref_adc_set bit[7] --> vref_adc	voltage

Table 12. Analog Monitor Settings.

16.3 Current Monitor

Various bias currents internal to LArPix-v3 can be observed using the current monitor pad (I_MONITOR). This pad is not available in the packaged version of LArPix-v3. The currents are sourced by LArPix-v3 so if monitoring a current there must be a resistor to ground from the I_MONITOR pad. The bias currents available for monitoring are listed in Table 13.

Configuration name	Bank Monitored	Signal Monitored	Voltage / Current
current_monitor_bank0[3:0]	0	bit[3] --> ibias_csa bit[2] --> ibias_buffer bit[1] --> ibias_comp bit[0] --> ibias_tdac	current
current_monitor_bank1[3:0]	1	bit[3] --> ibias_csa bit[2] --> ibias_buffer bit[1] --> ibias_comp bit[0] --> ibias_tdac	current
current_monitor_bank2[3:0]	2	bit[3] --> ibias_csa bit[2] --> ibias_buffer bit[1] --> ibias_comp bit[0] --> ibias_tdac	current
current_monitor_bank3[3:0]	3	bit[3] --> ibias_csa bit[2] --> ibias_buffer bit[1] --> ibias_comp bit[0] --> ibias_tdac	current

Table 13. Current Monitor settings.

16.4 Digital Monitor

The digital monitor is used to observe various internal digital signals at the DIG_MONITOR pin. Many of the signals are per-channel so to operate the digital monitor it must first be enabled, and then the channel as well as the signal selected.

To enable the digital monitor, set **digital_monitor_enable** to 1. The selected channel is chosen through **digital_monitor_chan[5:0]**. The signal to monitor can be determined by **digital_monitor_select[3:0]**. The decoding of **digital_monitor_select[3:0]** is summarized in Table 14.

digital_monitor_select[3:0] value	Digital Monitor Output Signal
0	hit
1	comp
2	sample
3	csa_reset
4	triggered_natural
5	periodic_trigger
6	periodic_reset
7	external_trigger
8	cross_trigger
9	reset_n_config_sync
10	sync_timestamp

Table 14. Digital Monitor Settings.

17 Biasing

The bias points of many of the amplifiers in LArPix-v3 are programmable. All trimming currents reference the Primary bias current.

17.1 Primary Bias

The Primary bias current is generated by an internal vt-referenced current source that is independent of the power supply voltage to first order. The current is trimmed using **ref_current_trim[4:0]**. The current is monitored through the I_MONITOR pad (LArPix-v3 sources the current). Adjust **ref_current_trim[4:0]** until the readback current i_readback is equal to 20 µA. The internal Primary current i_ref is equal to 1.5 * i_readback.

At **300 K**, the primary current i_ref is:

$$i_{ref} = 24.5\ \mu A + 0.5\ \mu A * ref_current_trim[4:0]$$

At **88 K**, the primary_current i_ref is:

$$i_{ref} = 22.75\ \mu A + 0.4\ \mu A * ref_current_trim[4:0]$$

17.2 Channel Biasing

Various channel blocks can have their bias points set independently. All the programming settings are referenced to the Primary internal bias current i_ref. The bias programming is summarized in Table 15.

Configuration Name	Block Name	Bias Transfer Function LSB
ibias_tdac[3:0]	Threshold DAC	LSB = i_ref/30
ibias_comp[3:0]	Discriminator	LSB = i_ref/30
ibias_buffer[3:0]	A/B Buffer	LSB = i_ref/10
ibias_csa[3:0]	CSA	LSB = 4*i_ref/25
ibias_vref_buffer[3:0]	VREF Buffer	LSB = i_ref/15 (control is active low, so ibias = (15-code)*i_ref/15))

ibias_vcm_buffer[3:0]	VCM Buffer	LSB = 2*i_ref/15 (control is active low, so ibias = (15-code)*2*i_ref/15))
ibias_tpulse[3:0]	CSA Pulser	LSB = i_ref/10

Table 15. Bias Current Programming.

17.3 ADC Delay

The asynchronous ADC includes an internal delay. This delay can be adjusted by increasing the value in **adc_ibias_delay**. More has been mentioned on this in Section 2.2.8.

18 Additional Modes and Settings

LArPix-v3 includes a number of functional settings and modes (for both normal operation and test) that are accessible through the Configuration Register Map.

18.1 Front-End Gain

The analog front end currently only has a one gain option (since the low-gain mode included in versions up to v2c and before remained unused and suspected to add noise). The nominal front-end gain is 4 μV/e-.

18.2 Front-End Bypass (to test ADC)

To enable the independent evaluation of the channel ADC, the front-end can be bypassed. Bypassing the front-end is a two-step process. First, the bypass channel must be enabled by setting **csa_bypass_enable** to 1. Then, the channels to bypass (i.e., the channels to which the test voltage pin should be applied) are selected by setting each bit corresponding to each channel of **csa_bypass_select[31:0]**. For example, to connect the test voltage pin (ADC_TEST_IN) to the ADCs in channel 31 and in channels 0 through 4,

csa_bypass_select[31:0] should be set to 0x8000001F. In this mode either a pulse on the external trigger pin or a test pulse of sufficient amplitude to cause a discriminator hit is required to initiate a conversion.

18.3 Power Supply Bypass Select

LArPix-v3 includes power supply bypassing in the analog front end to reduce transient interference. This bypassing can be disabled by the user if it causes ringing in the power supply network. To enable power supply bypassing, set **bypass_caps_enable** to 1. To disable, set it to 0. The bypassing is disabled by default.

18.4 Analog Test Pulse

LArPix-v3 includes an analog test pulse generator which can be used to characterize the channels. A conceptual diagram of the test pulse generator is shown in Figure 10.

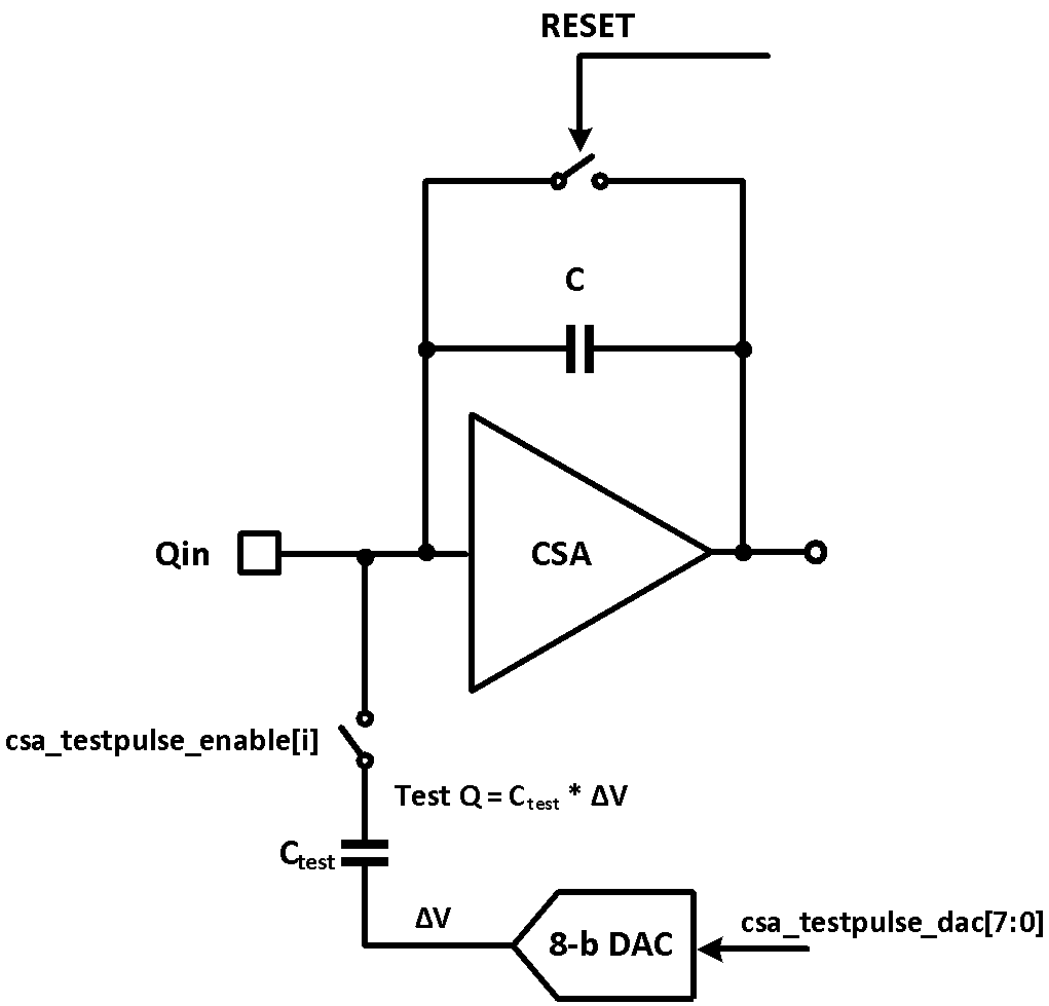


Figure 10. Internal analog test concept.

A channel-under-test is selected by closing the switch labeled **csa_testpulse_enable[i]**. The 8-b test DAC is then updated by changing the **csa_testpulse_dac[7:0]** word. This creates a voltage step ΔV that is coupled to the input of the analog front end by the test capacitor C_{test}. The voltage response of the channel can then be interpreted to extract the gain of the channel. In LArPix, C_{test} is approximately 40 fF. Because C_{test} is approximately 0.1% the size of the expected input capacitance of the channel, the input capacitance will not affect the input charge and the CSA analog output will be

$$V_{out} = \frac{\Delta V C_{test}}{C}$$

where C is the feedback capacitance of the CSA.

The command **csa_testpulse_enable[63:0]** is active low. Therefore, to create a test pulse, first select the channels of interest by setting their corresponding bits in **csa_testpulse_enable[63:0]** to zero**.** Then, update the test DAC by setting the desired value to **csa_testpulse_dac[7:0]**.

The LSB of the test pulse DAC is given by the analog power supply voltage divided by 256. So, if the **csa_testpulse_dac[7:0]** is 128 and the analog power supply is 1.8 V, ΔV would be (1.8V/256) * 128 = 900 mV.

18.5 Sample Stretching

The ADC in LArPix-v3 works by sampling the analog input onto the ADC capacitance and then applying the successive approximation (binary search) algorithm to converge on the closet digital representation of the sampled analog signal. The ADC capacitance is significant and therefore multiple clock cycles may be required to adequately settle depending on the system clock rate. The configuration word **adc_hold_delay[3:0]** determines how many system clock periods the ADC will spend sampling the data. For example, setting **adc_hold_delay[3:0]** to 0xF will result in 15 system clock periods allocated to sampling. Be advised that this will affect channel throughput. Settings zero and one both correspond to one system clock period for sampling.

19 Data Format

The following data is included in each 64-bit event:

1. 2-bit packet declaration (is it a data packet or a configuration packet?)
2. 8-bit chip ID (allowing up to 256 chips to be addressed as a unit)
3. 6-bit channel ID (allowing up to a 64-channel ASIC)
4. 30-bit timestamp
5. 10-bit data payload word
6. Trigger Type (Normal, External, Cross, or Periodic)
7. FIFO half full and FIFO full flags (both shared FIFO and local FIFO)
8. Marker to indicate if a packet is upstream or downstream
9. Parity Bit (LArPix-v3 implements odd parity).

In addition, several MSBs of the timestamp can be replaced with the current FIFO depth, if desired.

An event is defined as either a hit in any channel or a configuration request. Because the time between when a hit occurs and when it is transmitted off chip depends on a variety of factors, it is not deterministic. Therefore, LArPix-v3 includes a 32-bit timestamp. Assuming a 10 MHz system clock, for instance, a 32-bit time stamp will cover a range of over seven minutes. Additional timestamping can be implemented at the FPGA level.

19.1 Data Packet

The LArPix-v3 event output format (data mode) is shown in Table 16. This format is operative when the Packet Declaration bits are 01.

Bit Range	Contents	Comment
[1:0]	Packet Declaration	00: not used **01: data** 10: configuration write 11: configuration read
[9:2]	Chip ID	Each chip in a Hydra Network needs a unique ID
[15:10]	Channel ID	64-channel chip supported
[43:16]	Time Stamp	28-bit time stamp
[44]	Reset Sample Flag	If in CDS mode, set to 1 for reset sample, 0 for data samples
[45]	CDS Flag	0: no correlated double-sampling 1: correlated double-sampling
[55:46]	Data Word	10-bit ADC data word

[57:56]	Trigger Type	00: Normal, 01: External, 10: Cross, 11: Periodic
[59:58]	Local FIFO Status	Bit 58: FIFO half full, Bit 59: FIFO full
[61:60]	Shared FIFO Status	Bit 60: FIFO half full, Bit 61: FIFO full
[62]	Downstream Marker	0: Upstream Packet, 1: Downstream Packet
[63]	Parity Bit	0 if number of 1s in bits [62:0] is odd (odd parity)

Table 16. Structure of LArPix-v3 output word (data mode).

If the **enable_fifo_diagnostics** bit is set to 1, then bits [43:32] will contain the number of events sitting in the shared FIFO of the chip from which the event is received at the time the event was registered.

If the **enable_local_fifo_diagnostics** is set to 1, Bits [31:28] will contain number of events sitting in the given channel's local FIFO at the time the event was registered.

If the **mark_first_packet** bit is set to 1, then the MSB of the timestamp with be set to '1' for the first packet, and '0' for other packets in a burst.

19.2 Configuration Packet

LArPix-v3 is programmed by setting the Chip ID, Register Map Address, and Register Map Data to the desired values, setting the Packet Declaration to 10 and then sending the word to LArPix-v3. Setting the Chip ID and Register Map Address to the desired values and then setting the Packet Declaration to 11 will return the value of the Register Map Data for the chip and address identified. The LArPix-v3 event output format (configuration mode) is shown in Table 17. Note that bit[1] = 1 indicates a configuration packet, and given a configuration packet, bit[0] = 0 indicates write and bit[0] = 1 indicates read.

Bit Range	**Contents**	**Comment**
[1:0]	Packet declaration	00: not used 01: data

		10: configuration write **11: configuration read**
[9:2]	Chip ID	Each chip in a Hydra Network needs a unique ID
[17:10]	Register Map Address	
[25:18]	Register Map Data	
[57:26]	Magic Number	32-bit magic number must be present to validate transaction. Magic number: 0x89_50_4E_47[4]
[61:58]	All zeros	
[62]	Downstream Marker	0: Upstream Packet, 1: Downstream Packet
[63]	Parity Bit	0 if number of 1s in bits [62:0] is odd (odd parity)

Table 17. Structure of LArPix-v3 output word (configuration mode).

Bits [57:26] define a 32-bit magic number used by LArPix-v3 to protect the configuration from random packets. If the magic number is not included correctly then LArPix-v3 will ignore the configuration request. The required value of the magic number is specified in Table 17.

[4] The magic number used by LArPix-v3 is the same magic number used to identify PNG files.